

# Logistic Regression:

from Marginal/Ordinary  
to Random-Effects

# Marginal and Individual Probabilities

- Marginal (ordinary) logistic regression models the overall (population-averaged) probabilities
- Random effects logistic regression models the individual (subject-specific) probabilities

# Marginal (ordinary) logistic regression

- **Data:**  $(Y_{ij}, x_{ij1}, \dots, x_{ij,p-1})$  for  $i = 1, \dots, m, j = 1, \dots, n_i$  where  $Y_{ij} \in \{0, 1\}$  are binary.
- **Marginal (ordinary) logistic regression:**
  - Model the response  $Y$  as Bernoulli distribution:  
 $P(Y_{ij} = 1) = \pi_{ij}$  and  $P(Y_{ij} = 0) = 1 - \pi_{ij}$
  - Link model parameters to predictor  $X$  variables:  
$$\log \frac{\pi_{ij}}{1 - \pi_{ij}} = \beta_0 + \beta_1 x_{ij1} + \beta_2 x_{ij2} + \dots + \beta_{p-1} x_{ij,p-1}$$

# R code for GLM

```
glm(Y ~ X1+ ...+Xp, data = data0, family = binomial)
```

- In general, this is a convex optimization problem, and the maximum likelihood estimation (MLE) usually exists.
- When it has a warning message, it often means that there is a perfect fit!
  - In such a case, you might want to remove some predictor X variables, depending on the domain knowledge.

# Random Intercepts logistic regression

- **Data:**  $(Y_{ij}, x_{ij1}, \dots, x_{ij,p-1})$  for  $i = 1, \dots, m, j = 1, \dots, n_i$  where  $Y_{ij} \in \{0, 1\}$  are binary.
- **Random intercepts logistic regression:**
  - Model the response  $Y$  as Bernoulli distribution:  
 $P(Y_{ij} = 1) = \pi_{ij}$  and  $P(Y_{ij} = 0) = 1 - \pi_{ij}$
  - Link model parameters to predictor  $X$  variables:  
$$\log \frac{\pi_{ij}}{1 - \pi_{ij}} = \beta_0 + \beta_1 x_{ij1} + \beta_2 x_{ij2} + \dots + \beta_{p-1} x_{ij,p-1} + u_i$$
where the random-effects  $u_i$  are iid  $N(0, \sigma_u^2)$ .

# R codes

- In “lme4” R package:

`glmer(Y ~ X1+ ...+Xp + (1 | ID), data = data0, family = binomial)`

- Sometimes the default setting might not converge, and you can check

- Is it perfect fitting?

- Maybe we should remove some X variables!

- Does the numerical optimization algorithm converge?

- Maybe we need to change some settings:

`glmer(Y ~ X1+ ...+Xp + (1 | ID), data = data0, family = binomial, control = glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 100000)) )`

# A Simple case

- **Data:**  $(Y_{ij}, x_{ij})$  for  $i = 1, \dots, m$  where  $Y_{ij} \in \{0, 1\}$ ,  $x_{ij} \in \{0, 1\}$

**Example:**

|       | ID | Y | X |
|-------|----|---|---|
| [1,]  | 1  | 0 | 0 |
| [2,]  | 1  | 0 | 0 |
| [3,]  | 1  | 1 | 0 |
| [4,]  | 1  | 0 | 1 |
| [5,]  | 2  | 0 | 0 |
| [6,]  | 2  | 1 | 1 |
| ..... |    |   |   |

**The data can be summarized as m 2x2 table (this is typical in biostatistics)**

$$n_i = a_i + b_i + c_i + d_i$$

|              | $Y_{ij} = 1$ | $Y_{ij} = 0$ | Sum         |
|--------------|--------------|--------------|-------------|
| $x_{ij} = 1$ | $a_i$        | $b_i$        | $a_i + b_i$ |
| $x_{ij} = 0$ | $c_i$        | $d_i$        | $c_i + d_i$ |

# A Simple case

- The Simple case for Random-Intercepts Logistic Regression Model:

➤  $P(Y_{ij} = 1) = \pi_{ij}, \quad P(Y_{ij} = 0) = 1 - \pi_{ij}$

➤  $\log \frac{\pi_{ij}}{1-\pi_{ij}} = \beta_0 + \beta_1 x_{ij} + u_i$  where  $u_i \sim N(0, \sigma_u^2)$

- The likelihood function is

$$\begin{aligned} L(\boldsymbol{\beta}) &= \prod_{i=1}^m f(Y_{i1}, \dots, Y_{i,n_i}) = \prod_{i=1}^m \int_{-\infty}^{\infty} f(Y_{i1}, \dots, Y_{i,n_i} | u_i) f(u_i) du_i \\ &= \prod_{i=1}^m \int_{-\infty}^{\infty} \prod_{j=1}^{n_i} \frac{e^{Y_{ij}(\beta_0 + \beta_1 x_{ij} + u_i)}}{1 + e^{\beta_0 + \beta_1 x_{ij} + u_i}} \frac{1}{\sqrt{2\pi} \sigma_b} e^{-\frac{1}{2\sigma_u^2} u_i^2} du_i \end{aligned}$$



# Likelihood function: Further Simplification

- The likelihood function is

$$\begin{aligned} L(\boldsymbol{\beta}) &= \prod_{i=1}^m \int_{-\infty}^{\infty} \prod_{j=1}^{n_i} \frac{e^{Y_{ij}(\beta_0 + \beta_1 x_{ij} + u_i)}}{1 + e^{\beta_0 + \beta_1 x_{ij} + u_i}} \frac{1}{\sqrt{2\pi} \sigma_u} e^{-\frac{1}{2\sigma_u^2} u_i^2} du_i \\ &= \prod_{i=1}^m \int_{-\infty}^{\infty} \frac{e^{(\beta_0 + u_i)(a_i + c_i) + \beta_1 a_i}}{(1 + e^{\beta_0 + \beta_1 + u_i})^{a_i + b_i} (1 + e^{\beta_0 + u_i})^{c_i + d_i}} \frac{1}{\sqrt{2\pi} \sigma_u} e^{-\frac{1}{2\sigma_u^2} u_i^2} du_i \end{aligned}$$

- Unfortunately, there are no explicit mathematical solutions, and thus we need to rely on numerical optimization algorithms to solve it.

# Default Optimization Algorithm: Laplace Approximation

## Key Idea of Laplace Approximation:

- Suppose  $h(u)$  achieves a local maximum at  $u_0$  and thus  $h'(u_0) = 0$  and by Taylor's series expansion:

$$\begin{aligned} h(u) &\approx h(u_0) + h'(u_0)(u - u_0) + \frac{1}{2}h''(u_0)(u - u_0)^2 \\ &= h(u_0) + \frac{1}{2}h''(u_0)(u - u_0)^2 \end{aligned}$$

- Thus

$$\begin{aligned} \int_a^b e^{h(u)} du &\approx \int_a^b e^{h(u_0) + \frac{1}{2}h''(u_0)(u - u_0)^2} du \\ &= (\sqrt{2\pi} \sigma^*) e^{h(u_0)} \int_a^b \frac{1}{\sqrt{2\pi} \sigma^*} e^{-\frac{1}{2\sigma^{*2}}(u - u_0)^2} du \quad \text{with } \sigma^{*2} = -\frac{1}{h''(u_0)} \end{aligned}$$

where the integrand is just the pdf of  $N(u_0, \sigma^2)$ .

# Class Exercise

- Conduct the marginal model
    - Interpret the relevant estimates
  - Conduct the random-intercept model
    - Interpret the the relevant estimates
  - Describe the differences between the two modelling approaches
- ❖ R code is very different so pay attend in recitation!

# Administrative Stuff

- **Assignment 4 – due at Gradescope on April 10**  
please start sooner rather than later.
- **Assignment 5 – will vote for one of the following three options in class.**
  - a) It will be due April 17 in the standard Assignment format
  - b) It will be due April 17 in the in-class Lab format
  - c) It will be canceled/deleted

**In any case, we hope to allow you to have time to focus on your class project (the presentation files due noon on April 24).**

# Next Class...

- Hybrid model
  - How do these models differ from linear and logistic models?
  - Interpretation of coefficients
  - Things to consider
- General Estimating Equation (GEE)
  - Difference between this and random effects modelling
  - Linear models
  - Logistic models
  - Other models

Questions ?